

Fourier Series

Periodic fn

A fn is said to be periodic if the value of fn repeats after a fixed interval (α) i.e.

if $f(x)$ is a fn, then $f(x+\alpha) = f(x)$

Ex: $\sin x$, $\cos x$ are periodic fn with $\alpha = 2\pi$.

In a fourier series of a fn $f(x)$ of $\alpha = 2l$ in the interval $(\alpha, \alpha+2l)$ is given by $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) +$

$\sum_{n=1}^{\infty} \frac{\sin\left(\frac{n\pi x}{l}\right)}{l} \cdot b_n$ where $a_n =$

$$a_0 = \frac{1}{l} \int_{\alpha}^{\alpha+2l} f(x) dx$$

$$a_n = \frac{1}{l} \int_{\alpha}^{\alpha+2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$b_n = \frac{1}{l} \int_{\alpha}^{\alpha+2l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$a_n = 0$$

$$b_n = \frac{1}{l} \int_{\alpha}^{\alpha+2l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin(nx) dx$$

$$= \frac{1}{\pi} \left[-\frac{x \cos nx}{n} + \frac{\sin nx}{n^2} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[\left(-\frac{\pi \cos(n\pi)}{n} + \frac{\sin(n\pi)}{n^2} \right) - \left(-\frac{\pi \cos(n-\pi)}{n} + \frac{\sin(n-\pi)}{n^2} \right) \right]$$

$$= \frac{1}{\pi} \left[-\frac{\pi \cos(n\pi)}{n} + \frac{\sin(n\pi)}{n^2} - \frac{\pi \cos(n-\pi)}{n} + \frac{\sin(n-\pi)}{n^2} \right]$$

$$b_n = -\frac{2 \cos(n\pi)}{n} = -\frac{2(-1)^n}{n}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$= 0 + \sum_{n=1}^{\infty} 0 \cos\left(\frac{n\pi x}{\pi}\right) + \sum_{n=1}^{\infty} \frac{-2}{n} (-1)^n \sin(nx)$$

$$f(x) = \sum_{n=1}^{\infty} \frac{-2(-1)^n}{n} \sin(nx)$$

$$2. f(x) = |x|, \text{ interval } [-\pi, \pi]$$

$$\text{Given: } (\alpha, \alpha+2l) = [-\pi, \pi]$$

$$\alpha = -\pi$$

$$\alpha + 2l = \pi$$

$$-\pi + 2l = \pi$$

$$l = \pi$$

$$f(x) = |x|, \text{ we have } f(x) = \begin{cases} x & 0 \leq x \leq \pi \\ -x & -\pi \leq x \leq 0 \end{cases}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$a_0 = \frac{1}{l} \int_{-\pi}^{\pi} |x| dx = \frac{1}{\pi} \left(\int_{-\pi}^0 -x dx + \int_0^{\pi} x dx \right)$$

$$= \frac{1}{\pi} \left(\left[-\frac{x^2}{2} \right]_{-\pi}^0 + \left[\frac{x^2}{2} \right]_0^{\pi} \right)$$

$$= \frac{1}{\pi} \left(+\frac{\pi^2}{2} + \frac{\pi^2}{2} \right) = \frac{\pi}{2}$$

where a_0, a_n, b_n are called Euler's formulae

NOTE: If $f(x)$ is a fn, then we say that it is an even function if $f(-x) = f(x)$
Ex: $\cos x$

It is said to be odd fn, if $f(-x) = -f(x)$
Ex: $\sin x$

NOTE: $\int e^{ax} \cos bx \, dx = \frac{e^{ax}}{a^2 + b^2} [a \cos bx + b \sin bx]$

$\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx]$

Bernoulli's

$\int u^m v = u \int v - u' \int \int v + u'' \int \int \int v - u''' \int \int \int \int v + \dots$

$\cos(n\pi) = (-1)^n$

$\sin(n\pi) = 0$

Find F.S. of fn

1. $f(x) = x$ in the interval $(-\pi, \pi)$

→ Given: $(\alpha, \alpha + 2\ell) = (-\pi, \pi)$

$\alpha = -\pi$

$\alpha + 2\ell = \pi$

$-\pi + 2\ell = \pi$

$\ell = \pi$

$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{\ell}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{\ell}\right)$

$a_0 = \frac{1}{\ell} \int_{\alpha}^{\alpha+2\ell} f(x) \, dx = \frac{1}{\pi} \int_{-\pi}^{\pi} x \, dx$

$a_0 = \frac{1}{\pi} \left[\frac{x^2}{2} \right]_{-\pi}^{\pi} = \frac{1}{\pi} \left[\frac{\pi^2}{2} - \frac{\pi^2}{2} \right]$

$a_0 = 0$

$a_n = \frac{1}{\ell} \int_{\alpha}^{\alpha+2\ell} f(x) \cos\left(\frac{n\pi x}{\ell}\right) \, dx = \frac{1}{\pi} \int_{-\pi}^{\pi} x \cos\left(\frac{n\pi x}{\pi}\right) \, dx$

$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \cos nx \, dx = \frac{1}{\pi} \left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right]_{-\pi}^{\pi}$

$= \frac{1}{\pi} \left[\frac{\pi \sin(n\pi)}{n} + \frac{\cos n\pi}{n^2} + \frac{\pi \sin(-n\pi)}{n} + \frac{\pi \cos(n\pi)}{n^2} \right]$

$$\begin{aligned}
 a_n &= \frac{1}{L} \int_{-\pi}^{\pi} f(x) \cos\left(\frac{n\pi x}{L}\right) dx \\
 &= \frac{1}{\pi} \int_{-\pi}^{\pi} |x| \cos\left(\frac{n\pi x}{\pi}\right) dx \\
 &= \frac{1}{\pi} \left(\int_{-\pi}^0 -x \cos nx \, dx + \int_0^{\pi} x \cos nx \, dx \right) \\
 &= \frac{1}{\pi} \left(\left[-x \frac{\sin nx}{n} + \frac{\cos nx}{n^2} \right]_{-\pi}^0 + \left[x \frac{\sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^{\pi} \right) \\
 &= \frac{1}{\pi} \left(\left[-\frac{1}{n^2} - \frac{(-1)^n}{n^2} \right] + \left[\frac{(-1)^n}{n^2} - \frac{1}{n^2} \right] \right) \\
 &= \frac{1}{\pi} \left(-\frac{1}{n^2} + \frac{(-1)^n}{n^2} + \frac{(-1)^n}{n^2} - \frac{1}{n^2} \right) \\
 &= \frac{1}{\pi} \left(\frac{-2}{n^2} + \frac{2(-1)^n}{n^2} \right) = \frac{-2 + 2(-1)^n}{\pi n^2} \\
 \underline{a_n} &= \underline{\frac{2((-1)^n - 1)}{\pi n^2}}
 \end{aligned}$$

$$\begin{aligned}
 b_n &= \frac{1}{L} \int_{-\pi}^{\pi} f(x) \sin\left(\frac{n\pi x}{L}\right) dx \\
 &= \frac{1}{\pi} \int_{-\pi}^{\pi} |x| \sin(nx) \, dx \\
 &= \frac{1}{\pi} \left(\int_{-\pi}^0 -x \sin(nx) \, dx + \int_0^{\pi} x \sin(nx) \, dx \right) \\
 &= \frac{1}{\pi} \left(\left[+x \frac{\cos nx}{n} + \frac{\sin nx}{n^2} \right]_{-\pi}^0 + \left[-x \frac{\cos nx}{n} + \frac{\sin nx}{n^2} \right]_0^{\pi} \right) \\
 &= \frac{1}{\pi} \left(\left[+\frac{\pi(-1)^n}{n} + \frac{(-1)^n}{n^2} \right] + \left[-\frac{\pi(-1)^n}{n} + \frac{(-1)^n}{n^2} \right] \right) \\
 &= 0
 \end{aligned}$$

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2((-1)^n - 1) \cos nx}{\pi n^2}$$

NOTE: If $f(x)$ is a piecewise-continuous fn (dis) in $[-\pi, \pi]$ i.e. $f(x) = \begin{cases} \phi(x) & 0 \leq x < \pi \\ \psi(x) & -\pi \leq x < 0 \end{cases}$ then $f(x)$ is even $f(x)$ if $\phi(-x) = \psi(x)$ & $f(x)$ is odd if $\phi(x) = -\psi(x)$

16/4

$$1. f(x) = e^{-x}$$

$$\Rightarrow \text{we have F.C.T, } F_c(s) = \int_0^{\infty} f(x) \cos sx \, dx$$

$$= \int_0^{\infty} e^{-x} \cos sx \, dx$$

$$\text{NOTE: } \int_0^{\infty} e^{ax} \cos bx \, dx = \frac{e^{ax}}{a^2+b^2} [a \cos bx + b \sin bx]$$

$$\text{② } \int_0^{\infty} e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2+b^2} [a \sin bx - b \cos bx]$$

$$\Rightarrow \int_0^{\infty} \left[\frac{e^{-x}}{(-1)^2 + s^2} [-\cos sx + s \sin sx] \right]_0^{\infty}$$

$$= \frac{1}{s^2+1} \left[e^{-x} (-\cos sx + s \sin sx) \right]_0^{\infty}$$

$$= \frac{1}{s^2+1} [e^{-\infty} (-\cos s\infty + s \sin s\infty) - e^{-0} (-\cos 0 + s \sin 0)]$$

$$= \frac{1}{s^2+1} [0 - 1(-1+0)] = \frac{1}{s^2+1}$$

$$F_c(s) = \int_0^{\infty} f(x) \sin sx \, dx$$

$$= \int_0^{\infty} e^{-x} \sin sx \, dx = \left[\frac{e^{-x}}{(-1)^2 + s^2} [-\sin sx - s \cos sx] \right]_0^{\infty}$$

$$= \frac{1}{s^2+1} [-e^{-x} (\sin sx + s \cos sx)]_0^{\infty}$$

$$= \frac{1}{s^2+1} [-e^{-\infty} (\sin s\infty + s \cos s\infty) + e^{-0} (\sin 0 + s \cos 0)]$$

$$= \frac{1}{s^2+1} [1(0+s)] = \frac{s}{s^2+1}$$

$$2. f(x) = \begin{cases} x, & 0 < x < 1 \\ 0, & x > 1 \end{cases}$$

$$F_c(s) = \int_0^{\infty} f(x) \cos sx \, dx = \int_0^1 x \cos sx \, dx$$

$$= \left[x \frac{\sin sx}{s} + \frac{\cos sx}{s^2} \right]_0^1 = \left[\frac{\sin(s)}{s} + \frac{\cos(s)}{s^2} \right]$$

$$= \frac{\sin(s)}{s} + \frac{\cos(s)}{s^2} - \frac{1}{s^2}$$

we have $F(f(x)) = \int_{-\infty}^{\infty} f(x) \cdot e^{isx} dx$

$$= \int_{-1}^1 1 \cdot e^{isx} dx$$

$$= \left[\frac{e^{isx}}{is} \right]_{-1}^1 = \frac{e^{is} - e^{-is}}{is}$$

$$= \frac{1}{s} \left[\frac{e^{is} - e^{-is}}{i} \right] = \frac{2 \sin s}{s}$$

2. $f(x) = \begin{cases} (1-x)^2, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$

For a given $f(x)$, we have $f(x) = (1-x)^2$ if $-1 \leq x \leq 1$ & $f(x) = 0$ otherwise

we have $F(f(x)) = \int_{-\infty}^{\infty} f(x) \cdot e^{isx} dx$

$$= \int_{-1}^1 (1-x)^2 e^{isx} dx$$

$$F(f(x)) = \int_{-1}^1 e^{isx} dx - \int_{-1}^1 x^2 e^{isx} dx$$

$$= \frac{e^{isx}}{is} \Big|_{-1}^1 - \int_{-1}^1 x^2 e^{isx} dx$$

$$= \frac{e^{is} - e^{-is}}{is} - \left[\frac{x^2 e^{isx}}{is} - \frac{2x e^{isx}}{i^2 s^2} - \frac{2e^{isx}}{i^3 s^3} \right]_{-1}^1$$

$$= \frac{e^{is} - e^{-is}}{is} - \left[\frac{e^{is} + 2e^{is} + 2e^{is}}{is^3} - \frac{e^{-is} - 2e^{-is} - 2e^{-is}}{is^3} \right]$$

$$= \frac{e^{is} - e^{-is}}{is} - \frac{e^{is} - e^{-is}}{is^3} + \frac{2e^{is} - 2e^{-is}}{s^2}$$

$$= -\frac{2}{s^2} \left[\frac{e^{is} + e^{-is}}{2} \right] + \frac{2}{is^3} \left[\frac{e^{is} - e^{-is}}{2} \right]$$

$$= -\frac{2}{s^2} (2 \cos s) + \frac{2}{is^3} (2i \sin s)$$

Find F.T of s & c .

If $f(x)$ is even, then $b_n = 0$ & if $f(x)$ is odd, then the terms a_0 & $a_n = 0$.

$$3. f(x) = \begin{cases} -k & -\pi \leq x < 0 \\ +k & 0 \leq x \leq \pi \end{cases}, f(x+2\pi) = f(x)$$

$$[0, 2\pi] = [\alpha, \alpha + 2\ell]$$

$$\alpha = 0, \pi$$

$$\ell = \pi$$

$$\therefore f(x) = \sum_{n=1}^{\infty} \frac{2k}{n\pi} (1 - (-1)^n) \sin(nx)$$

$$a_0 = \frac{1}{\ell} \int_{\alpha}^{\alpha+2\ell} f(x) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^0 -k dx + \int_0^{\pi} k dx \right]$$

$$= \frac{1}{\pi} [-\pi k + \pi k] = 0$$

$$a_n = \frac{1}{\ell} \int_{\alpha}^{\alpha+2\ell} f(x) \cos\left(\frac{n\pi x}{\ell}\right) dx = \frac{1}{\pi} \left[\int_{-\pi}^0 -k \cos\left(\frac{n\pi x}{\pi}\right) dx + \int_0^{\pi} k \cos\left(\frac{n\pi x}{\pi}\right) dx \right]$$

$$= \frac{1}{\pi} [-k(0) + k(0)] = 0$$

$$b_n = \frac{1}{\ell} \left[\int_{\alpha}^{\alpha+2\ell} f(x) \sin\left(\frac{n\pi x}{\ell}\right) dx \right] = \frac{1}{\pi} \left[\int_{-\pi}^0 -k \sin\left(\frac{n\pi x}{\pi}\right) dx + \int_0^{\pi} k \sin\left(\frac{n\pi x}{\pi}\right) dx \right]$$

$$= \frac{1}{\pi} \left[\frac{k}{n} \left(\cos(n\pi) - \cos(0) \right) + \frac{k}{n} \left(-\cos(n\pi) + \cos(0) \right) \right] = \frac{2k}{n\pi} (1 - (-1)^n)$$

$$4. f(x) = \begin{cases} 0 & -2 < x < 0 \\ 1 & 0 \leq x < 2 \end{cases}, f(x+4) = f(x)$$

$$[0, 4] = [\alpha, \alpha + 2\ell]$$

$$\alpha = 0$$

$$-0 + 2\ell = 4$$

$$\ell = 2$$

$$a_0 = \frac{1}{\ell} \int_{\alpha}^{\alpha+2\ell} f(x) dx = \frac{1}{2} \left[\int_0^2 0 dx + \int_2^4 1 dx \right]$$

$$a_0 = \frac{1}{2} [2(0) + 2] = 1$$

$$a_n = \frac{1}{\ell} \left[\int_{\alpha}^{\alpha+2\ell} f(x) \cos\left(\frac{n\pi x}{\ell}\right) dx \right] = \frac{1}{2} \left[\int_0^2 0 \cos\left(\frac{n\pi x}{2}\right) dx + \int_2^4 1 \cos\left(\frac{n\pi x}{2}\right) dx \right]$$

$$a_n = \frac{1}{2} \left[\frac{\sin\left(\frac{n\pi x}{2}\right) \cdot \frac{2}{n\pi}}{\frac{2}{n\pi}} \right]_2^4 = \frac{1}{n\pi} (\sin(2n\pi) - \sin(0)) = 0$$

$$5. \int_0^t \frac{\sin t}{t} dt = \frac{1}{s} \tan^{-1}\left(\frac{1}{s}\right)$$

$$6. \int_0^t e^{-t} \cos t \cdot dt$$

$$= \frac{1}{s} \left(\mathcal{L}\{e^{-t} \cos t\} \right) = \frac{1}{s} \left(\mathcal{L}\{\cos t\} \right)$$

$$= \frac{1}{s} \left(\frac{s}{s^2 + a^2} \right)_{s \rightarrow s+a} = \frac{1}{s} \left(\frac{s}{s^2 + 1} \right)$$

$$= \frac{1}{s} \left(\frac{s+1}{(s+1)^2 + 1} \right)$$

25/4 Inverse Laplace Transformation

Let $f(t)$ be a periodic fn. Then $\mathcal{L}\{f(t)\} = f(s)$

The Inverse LT. of $F(s)$ can be defined as $\mathcal{L}^{-1}\{F(s)\} = f(t)$ where $\mathcal{L}^{-1}$ is a Inverse Laplace transformation operator

$$\begin{aligned} \mathcal{L}^{-1}\{a/s\} &= a \\ \mathcal{L}^{-1}\{s/(s^2+a^2)\} &= \cos at \\ \mathcal{L}^{-1}\{a/(s^2+a^2)\} &= \sin at \\ \mathcal{L}^{-1}\{a/(s-a)\} &= e^{at} \\ \mathcal{L}^{-1}\{1/(s-a)\} &= e^{-at} \\ \mathcal{L}^{-1}\{1/(s+a)\} &= \cos hat \\ \mathcal{L}^{-1}\{s/(s^2-a^2)\} &= \cos hat \\ \mathcal{L}^{-1}\{a/(s^2-a^2)\} &= \sin hat \\ \mathcal{L}^{-1}\{1/s^{n+1}\} &= t^n/n! \\ \mathcal{L}^{-1}\{1/s^{n+1}\} &= t^n/F(n+1) \\ \mathcal{L}^{-1}\{F(s-a)\} &= e^{at} \mathcal{L}^{-1}\{F(s)\} \\ \mathcal{L}^{-1}\{F(s-a)\} &= e^{at} f(t) \end{aligned}$$

Find ILT.

$$\frac{1}{2s-5} \Rightarrow \mathcal{L}^{-1}\left(\frac{1}{2s-5}\right) = \mathcal{L}^{-1}\left(\frac{1}{2(s-\frac{5}{2})}\right)$$

$$= \frac{1}{2} \mathcal{L}^{-1}\left(\frac{1}{s-\frac{5}{2}}\right) = \frac{1}{2} e^{\frac{5}{2}t}$$

$$2. \mathcal{L}^{-1}\left(\frac{2s-5}{4s^2+2s}\right)$$

$$\begin{aligned}
 & \log s \Big|_s^\infty - \frac{1}{2} \log (s^2 + a) \Big|_s^\infty \\
 &= \log \left(\frac{s}{(s^2 + a)^{1/2}} \right) \Big|_s^\infty \\
 &= \lim_{s \rightarrow \infty} \log \left(\frac{s}{s(1 + \frac{a}{s^2})^{1/2}} \right) - \log \left(\frac{s}{(s^2 + a)^{1/2}} \right) \\
 &= \log \left(\frac{(s^2 + a)^{1/2}}{s} \right)
 \end{aligned}$$

$$4. L \left\{ \frac{e^{-t} \sin t}{t} \right\}$$

$$= L \left\{ \frac{\sin t}{t} \right\}_{s \rightarrow s+1}$$

$$= \int_s^\infty \frac{1}{s^2 + a^2} ds \Big|_{s \rightarrow s+1} = \int_{s+1}^\infty \frac{1}{(s+1)^2 + 1} ds$$

$$\Rightarrow \tan^{-1} \left(\frac{1}{s} \right) \Big|_{s \rightarrow s+1} = \tan^{-1} \left(\frac{1}{s+1} \right)$$

Find L.T.

1. $\sin 3t$

$$= L[\sin at] = a / (s^2 + a^2)$$

$$L[\sin 3t] = 3 / (s^2 + 9)$$

2. $\cos 3t \cdot \cos t = 1/2 [c3t + ct]$
 $L[\cos t \cdot \cos 3t] = 1/2 L[c3t + ct] = 1/2 [L[3t] + L[t]]$
 $= \frac{1}{2} \left[\frac{s}{s^2+9} + \frac{s}{s^2+1} \right]$

3. $\sin t \cdot \sin 3t = 1/2 [c4t - c2t]$
 $L[\sin t \cdot \sin 3t] = 1/2 [L[4t] - L[2t]]$
 $= \frac{1}{2} \left[\frac{s}{s^2+16} - \frac{s}{s^2+4} \right]$

4. $1 + \cosh 6t$

$$L[1/2] + 1/2 L[\cosh 6t] = \frac{1}{2s} + \frac{s}{2(s^2 - 36)}$$

11/11

5. $\sin 3t \cdot \cos 4t = 1/2 [s7t - st]$
 $L[s3t \cdot c4t] = 1/2 (L[s7t] - L[st])$
 $= \frac{1}{2} \left[\frac{7}{s^2+49} - \frac{1}{s^2+1} \right]$

6. $\cosh 3t = 1 + \cosh 6t$
 $L[\cosh 3t] = L[1/2] + 1/2 L[\cosh 6t]$
 $= \frac{1}{2s} + \frac{1}{2} \left[\frac{s}{s^2-36} \right]$

7. $\cos(a-b) = \cos a \cos b + \sin a \sin b$
 $L[\cos(a-b)] = \cos b \left(\frac{s}{s^2+a^2} \right) + \sin b \left(\frac{a}{s^2+a^2} \right)$

8. $e^{3t+1} = e^{3t} e^1$
 $L(e^{3t+1}) = e^1 \left(\frac{1}{s-3} \right)$

9. $t^3 + t^2 + 17t - 1 \Rightarrow L(t^3) + 1/2 L(t^2) + 17 L(t) - L(1)$
 $L(t^3 + t^2/2 + 17t - 1) = \frac{6}{s^4} + \frac{1}{s^3} + \frac{17}{s^2} - \frac{1}{s}$

10. $\sqrt{t} - \frac{1}{\sqrt{t}} = t^{1/2} - t^{-1/2}$
 $L(\sqrt{t} - \frac{1}{\sqrt{t}}) = L(t^{1/2}) - L(t^{-1/2})$
 $= \frac{\sqrt{\pi}}{2s^{3/2}} - \frac{\sqrt{\pi}}{s^{1/2}}$

$$= \frac{-4+s^2}{(s^2+4)^2} + \frac{1}{s+4}$$

$$6. L\left\{\frac{\sin t}{t}\right\}$$

$$\Rightarrow L\{t^{-1} \sin t\} = \int_0^{\infty} \frac{1}{s^2+1} ds$$

$$\int_0^{\infty} \tan^{-1}\left(\frac{1}{s}\right) ds$$

$$\begin{aligned} \text{NOTE: } \int_0^{\infty} \frac{1}{s^2+1} ds &= \tan^{-1}(\infty) - \tan^{-1}(0) \\ &= \frac{\pi}{2} - \tan^{-1}(0) \\ &= \cot^{-1}(0) = \tan^{-1}\left(\frac{1}{0}\right) \end{aligned}$$

22/4 Find Laplace Transformation

$$1. L\left\{\frac{\sin t}{t} + t \sin t\right\}$$

$$= \tan^{-1}\left(\frac{1}{s}\right) + (-1) \frac{d}{ds} \left(\frac{1}{s^2+1} \right) = \tan^{-1}\left(\frac{1}{s}\right) + \frac{2s}{(s^2+1)^2}$$

$$\tan^{-1}\left(\frac{1}{s}\right) + \frac{2s}{(s^2+1)^2}$$

$$2. L\left\{\frac{\cos at - \cos bt}{t}\right\}$$

$$= L\left\{\frac{\cos at}{t}\right\} - L\left\{\frac{\cos bt}{t}\right\}$$

$$= \int_0^{\infty} \frac{s}{s^2+a^2} ds - \int_0^{\infty} \frac{s}{s^2+b^2} ds$$

$$= \frac{1}{2} \left[\log(s^2+a^2) - \log(s^2+b^2) \right]_0^{\infty}$$

$$= \frac{1}{2} \left[\log\left(\frac{s^2+a^2}{s^2+b^2}\right) \right]_0^{\infty} = \log\left(\frac{s^2+b^2}{s^2+a^2}\right)^{1/2}$$

$$3. L\left\{\frac{1-\cos t}{t}\right\} = L\left\{\frac{1}{t}\right\} - L\left\{\frac{\cos t}{t}\right\}$$

$$= \int_0^{\infty} \frac{1}{s} ds - \int_0^{\infty} \frac{s}{s^2+1} ds$$

$$\begin{aligned}
 &= L^{-1}\left(\frac{2s}{4s^2-5^2}\right) - L^{-1}\left(\frac{s}{4s^2-5^2}\right) \\
 &= L^{-1}\left(\frac{2s}{4(s^2-5^2)}\right) - L^{-1}\left(\frac{s}{4(s^2-5^2)}\right) \\
 &= \frac{1}{2} L^{-1}\left(\frac{s}{s^2-(\frac{5}{2})^2}\right) - \frac{1}{4} L^{-1}\left(\frac{s}{s^2-(\frac{5}{2})^2}\right) \\
 &= \frac{1}{2} L^{-1}\left(\frac{s}{s^2-(5/2)^2}\right) - \frac{1}{2} L^{-1}\left(\frac{5/2}{s^2-(5/2)^2}\right) \\
 &= \frac{1}{2} \left[\cos \frac{5t}{2} - \sin \frac{5t}{2} \right]
 \end{aligned}$$

$$\begin{aligned}
 3 \frac{4s-9}{9-s^2} &\Rightarrow L^{-1}\left(\frac{9}{s^2-9}\right) - L^{-1}\left(\frac{49}{s^2-9}\right) \\
 &= L^{-1}\left(\frac{3^2}{s^2-3^2}\right) - 4 L^{-1}\left(\frac{s}{s^2-3^2}\right) \\
 &= 3 \sin 3t - 4 \cos 3t
 \end{aligned}$$

$$\begin{aligned}
 &= L^{-1}\left(\frac{s+s-3}{s^2-6s+9+4}\right) \\
 &= L^{-1}\left(\frac{s+s}{s^2-6s+13}\right) + L^{-1}\left(\frac{-3}{s^2-6s+13}\right) \\
 &= L^{-1}\left(\frac{s-3}{(s-3)^2+2^2}\right) + 4 L^{-1}\left(\frac{2}{(s-3)^2+2^2}\right) \\
 &= e^{3t} \cos 3t + 4 e^{3t} \sin 2t \\
 &= L^{-1}(s^2-3s+4/s^3) \\
 &= L^{-1}\left(\frac{1}{s}\right) - L^{-1}\left(\frac{3}{s^2}\right) + L^{-1}\left(\frac{4}{s^3}\right) \\
 &= 1 - 3t + 4\left(\frac{t^2}{2}\right) \\
 &= 2t^2 - 3t + 1
 \end{aligned}$$

$$6. L^{-1}\left(\frac{1}{s(s-1)}\right)$$

$$= L^{-1}\left(\frac{1}{s(s-1)}\right) = \frac{1}{s(s-1)} = \frac{A(s-1) + B}{s(s-1)}$$

$$1 = A(s-1) + Bs$$

$$\text{put } s=0$$

$$-1 = A$$

$$\text{put } s=1$$

$$1 = B$$

$$\frac{1}{s(s-1)} = \frac{-(s-1) + s}{s(s-1)}$$

$$L^{-1}\left(\frac{1}{s(s-1)}\right) = L^{-1}\left(\frac{-1}{s} + \frac{1}{s-1}\right)$$

$$= -\frac{1}{s} + \frac{1}{s-1}$$

$$= -1 + e^t$$

$$\text{HW. } \frac{1}{(s+1)(s-2)}$$

$$\frac{3s + 5\sqrt{2}}{s^2 + 8}$$

17/4

Properties of LT1. If $\mathcal{L}\{f(t)\} = F(s)$ then

i) $\mathcal{L}\{e^{at}f(t)\} = \mathcal{L}\{f(t)\}$

ii) $\mathcal{L}\{e^{-at}f(t)\} = \mathcal{L}\{f(t)\}$

$\Rightarrow F(s-a)$ [s → s+a]
 $\Rightarrow F(s+a)$ [s → s-a]

2. $\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$

3. $\mathcal{L}\left\{\int_0^t f(t) dt\right\} = \frac{1}{s} F(s)$

4. $\mathcal{L}\left\{f\left(\frac{t}{s}\right)\right\} = \int_s^\infty F(s) \cdot ds$

21/4 Sums

1. $e^{-3t} [2 \cos 5t - 3 \sin 5t]$

$\Rightarrow 2e^{-3t} \cos 5t - 3e^{-3t} \sin 5t$

$\mathcal{L}\{2e^{-3t} \cos 5t\} - \mathcal{L}\{3e^{-3t} \sin 5t\}$

$2\mathcal{L}\{e^{-3t} \cos 5t\} - 3\mathcal{L}\{e^{-3t} \sin 5t\}$

$2\mathcal{L}\{\cos 5t\} - 3\mathcal{L}\{\sin 5t\}$

$\frac{2s}{s^2+25} - \frac{3(5)}{s^2+25}$

$$= \frac{2s+6}{s^2+9+6s+25} - \frac{15}{s^2+9+6s+25}$$

$$= \frac{2s-9}{s^2+6s+34}$$

$$2e^{-4t} t^{3/2}$$

$$\Rightarrow \mathcal{L}\{e^{-4t} t^{3/2}\} = \mathcal{L}\{t^{3/2}\} \quad s \rightarrow s+4$$

$$= \frac{\Gamma(5/2)}{s^{5/2}} = \frac{3/2 \sqrt{\pi}/2}{s^{5/2}}$$

$$= \left[\frac{3\sqrt{\pi}}{4s^{5/2}} \right]_{s=s+4} = \frac{3\sqrt{\pi}}{4(s+4)^{5/2}}$$

3. $(t+2)^2 e^t$

$$\Rightarrow \mathcal{L}\{e^t (t+2)^2\} = \mathcal{L}\{e^t t^2\} + \mathcal{L}\{e^t 4t\}$$

$$= \left[\frac{3}{s^3} \right]_{s \rightarrow s-1} + \left[\frac{4}{s^2} \right]_{s \rightarrow s-1} + \frac{4}{s-1}$$

Fourier Transforms

is a method that converts a function from the Time domain into Freq. domain

Ex: Music, where Time domain: Sound waves
Freq. domain: Pitch, Rhythm

F.T of fn $f(x)$ is:

$$F\{f(x)\} = - \int_{-\infty}^{\infty} f(x) \cdot e^{isx} \cdot dx = F(s)$$

i is imaginary no, s is freq. parameter.

Inverse F.T

It is defined as: $F\{F(s)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(s) \cdot e^{-isx} \cdot ds = f(x)$

Fourier sine & cosine Transform

$$F_c(s) = \int_0^{\infty} f(x) \cdot \cos Sx \cdot dx$$

$$F_s(s) = \int_0^{\infty} f(x) \cdot \sin Sx \cdot dx$$

I.F.T for sin & cos.

$$F^{-1}(F_c(s)) = \frac{2}{\pi} \int_0^{\infty} F_c(s) \cdot \cos Sx \cdot ds$$

$$F^{-1}(F_s(s)) = \frac{2}{\pi} \int_0^{\infty} F_s(s) \cdot \sin Sx \cdot ds$$

$\sin x: \frac{e^{ix} - e^{-ix}}{2i}$; $\cos x: \frac{e^{ix} + e^{-ix}}{2}$
NOTE: $|x| < a$ i.e. $-a < x < a$
 $|x| > a$ i.e. $-\infty \leq x \leq -a$ & $a \leq x \leq \infty$

Summ (Obtain F.T)

$$f(x) = \begin{cases} 1 & |x| < 1 \\ 0 & |x| > 1 \end{cases}$$

For a given $f(x)$, $f(x) = 1$, if $-1 \leq x \leq 1$
& $f(x) = 0$ other wise.

$$F_s(s) = \int_0^{\infty} f(x) \cdot \sin Sx \, dx = \int_0^{\infty} x \cos ax \, dx$$

$$= \left[-\frac{x \cos Sx}{S} + \frac{\sin Sx}{S^2} \right]_0^{\infty} = \left[-\frac{\cos(s)}{S} + \frac{\sin(s)}{S^2} \right]_0^{\infty}$$

Laplace Transform

If $f(t)$ is a real valued fn defined for all $t \geq 0$, then the Laplace transform of $f(t)$ defined by: $L\{f(t)\} = \int_0^{\infty} f(t) e^{-st} \, dt$

i.e. $= f(s)$ where s is freq. parameter & given by $s = \sigma + i\omega$.

let $f(t) = a$ (const), $L[a] = \int_0^{\infty} a e^{-st} \, dt$

$$= a \int_0^{\infty} e^{-st} \, dt = a \left[\frac{e^{-st}}{-s} \right]_0^{\infty} = \frac{a}{-s} \left[e^{-s\infty} - 1 \right]$$

$$= \frac{a}{-s} [0 - 1] = \frac{a}{s}$$

$$L[1] = a/s$$

$$L[\cosh at] = \frac{a}{s^2 - a^2}$$

$$L[\sinh at] = \frac{a}{s^2 + a^2}$$

$$L[\cos at] = \frac{s}{s^2 + a^2}$$

$$L[\sin at] = \frac{a}{s^2 + a^2}$$

$$L[e^{at}] = 1/s + a$$

$$L[e^{at}] = 1/s - a$$

$$L[t^n] = \Gamma(n+1) / s^{n+1}, \quad n \geq 0$$

$$\Gamma(n+1) = n!$$

$$\Rightarrow L[t^n] = n! / s^{n+1}, \quad n \geq 0$$

$$\Gamma(3/2) = \sqrt{\pi}/2$$

$$\Gamma(1/2) = \sqrt{\pi}$$

$$\cos A \cos B = 1/2 [C(A+B) + C(A-B)]$$

$$\sin A \sin B = 1/2 [C(A+B) - C(A-B)]$$

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A-B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\cos A \sin B = -1/2 [C(A+B) - C(A-B)]$$

$$\sin A \cos B = 1/2 [C(A+B) + C(A-B)]$$

$$\cosh^2 x = 1 + \frac{\cosh 2x}{2}$$

NOTE: $L[af(t)] = a L[f(t)]$

$$L[a + f(t)] = L[a] + L[f(t)]$$

$$= \frac{3}{(s-1)^3} + \frac{4}{(s-1)^2} + \frac{4}{s-1}$$

$$3e^{2t} + 4t^3 - 2\sin 3t + 3\cos 3t$$

$$\Rightarrow \mathcal{L}\{e^{2t}\} + 4\mathcal{L}\{t^3\} - 2\mathcal{L}\{\sin 3t\} + 3\mathcal{L}\{\cos 3t\}$$

$$\frac{1}{s-2} + \frac{4(3!)}{s^4} - \frac{2(3)}{s^2+9} + \frac{3(s)}{s^2+9}$$

$$= \frac{1}{s-2} + \frac{24}{s^4} - \frac{6}{s^2+9} + \frac{3s}{s^2+9}$$

$$4. \mathcal{L}\{t \sin at + t^5\}$$

$$\Rightarrow \mathcal{L}\{t \sin at\} + \mathcal{L}\{t^5\}$$

$$= (-1) \frac{d}{ds} \left(\frac{a}{s^2+a^2} \right) + \frac{5!}{s^6}$$

$$= -a \frac{d}{ds} \left(\frac{1}{s^2+a^2} \right) + \frac{120}{s^6}$$

$$= -a \left(\frac{-1}{(s^2+a^2)^2} (2s) \right) + \frac{120}{s^6}$$

$$= \frac{2as}{(s^2+a^2)^2} + \frac{120}{s^6}$$

$$s \mathcal{L}\{t \cos 2t\} + \mathcal{L}\{e^{-4t}\}$$

$$\Rightarrow \mathcal{L}\{t \cos 2t\} + \mathcal{L}\{e^{-4t}\}$$

$$= (-1) \frac{d}{ds} \left(\frac{s}{s^2+4} \right) + \frac{1}{s+4}$$

$$= (-1) \frac{d}{ds} \left(\frac{s}{s^2+4} \right) + \frac{1}{s+4}$$

$$= -\left(\frac{s(2s) - (s^2+4)}{(s^2+4)^2} \right) + \frac{1}{s+4} = \frac{-s^2+4+2s^2}{(s^2+4)^2} + \frac{1}{s+4}$$

$$b_n = \frac{1}{2} \int_{-\pi}^0 0 \sin\left(\frac{n\pi x}{2}\right) + \int_0^{\pi} 1 \sin\left(\frac{n\pi x}{2}\right) dx$$

$$= \frac{1}{2} \left[-\cos\left(\frac{n\pi x}{2}\right) \right]_0^{\pi} = \frac{1}{2} \left[-\cos\left(\frac{n\pi}{2}\right) + 1 \right]$$

$$= \frac{1}{n\pi} \left[-(-1)^n + 1 \right]$$

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{1}{n\pi} \left[-(-1)^n + 1 \right] \sin\left(\frac{n\pi x}{2}\right)$$

5. $f(x) = x^2, [-\pi, \pi]$

$\omega = \pi; 2+2L = \pi \Rightarrow L = \frac{\pi}{2}$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{1}{\pi} \left[\frac{x^3}{3} \right]_{-\pi}^{\pi} = \frac{1}{3\pi} \left[\pi^3 + \pi^3 \right] = \frac{2\pi^2}{3}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos nx dx = \frac{1}{\pi} \left[\frac{x^2 \sin nx}{n} + \frac{2x \cos nx}{n^2} - \frac{2 \sin nx}{n^3} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{2\pi(-1)^n}{n^2} + \frac{2\pi(-1)^n}{n^2} \right] = \frac{4(-1)^n}{n^2}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \sin nx dx = \frac{1}{\pi} \left[\frac{-x^2 \cos nx}{n} + \frac{2x \sin nx}{n^2} + \frac{2 \cos nx}{n^3} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[-\pi^2(-1)^n + 2(-1)^n + \pi^2(-1)^n - 2(-1)^n \right] = 0$$

$$b_n = 0$$

$$f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4(-1)^n}{n^2} \cos nx$$

Dirichlet's condition

We can obtain the Fourier series of a function $f(x)$ as: $\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$

only if it satisfies following conditions:

1. $f(x)$ is periodic, single valued & finite
2. $f(x)$ has at the most finite no. of maxima & minima
3. $f(x)$ has finite no. of discontinuities

Theorem:

If $f(t)$ is a periodic function with period T such that $f(t+nT) = f(t)$ for all values of t .

Prove that $L\{f(t)\} = \frac{1}{1-e^{-sT}} \int_0^{\infty} e^{-st} f(t) dt$

We have by the definition,

$$L\{f(t)\} = \int_0^{\infty} e^{-st} f(t) \cdot dt$$

let $t = u$, $dt = du$

$$\begin{aligned} L\{f(t)\} &= \int_0^{\infty} e^{-su} f(u) \cdot du \\ &= \int_0^T e^{-su} f(u) \cdot du + \int_T^{2T} e^{-su} f(u) du + \\ &\quad \dots + \int_{nT}^{(n+1)T} e^{-su} f(u) du + \dots \end{aligned}$$

$$L\{f(t)\} = \sum_{n=0}^{\infty} \int_{nT}^{(n+1)T} e^{-su} f(u) du \quad \text{--- (1)}$$

let $u = t + nT$, $du = dt$

When $u = nT$, $t = 0$

When $u = (n+1)T$, $t = T$

Since $f(u) = f(t+nT) = f(t)$,

Substituting in (1)

$$L\{f(t+nT)\} = \sum_{n=0}^{\infty} \int_0^T e^{-s(t+nT)} f(t+nT) dt$$

$$L\{f(t)\} = \sum_{n=0}^{\infty} \int_0^T e^{-st} e^{-snT} f(t) dt$$

$$L\{f(t)\} = \sum_{n=0}^{\infty} e^{-snT} \int_0^T e^{-st} f(t) dt$$

$$L\{f(t)\} = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) dt \quad \left[e^{-snT} = \frac{1}{1-e^{-sT}} \right]$$

by geometric series